2021 F=ma Exam: Problem 16

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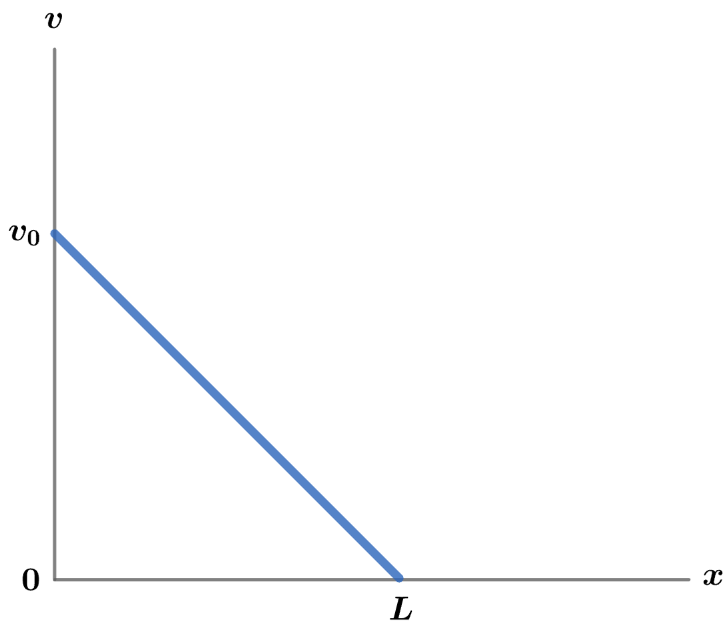

We have

$$v = \frac{dx}{dt} = v_0 - \frac{v_0}{L}x$$
$$\frac{dx}{1 - x/L} = v_0\,dt$$

Integrating,

$$\int_0^x \frac{dx'}{1 - x'/L} = \int_0^t v_0\,dt'$$
$$-L\log(1 - x/L) = v_0 t$$
$$1 - \frac{x}{L} = e^{-\frac{v_0 t}{L}}$$

We have

$$x(t) = L(1 - e^{-\frac{v_0 t}{L}})$$
$$v(t) = v_0 e^{-\frac{v_0 t}{L}}$$

Thus, with the caveat that the particle's velocity asymptotically goes to zero as $x \to L$, the answer is $\boxed{\text{A}}$.